

EMPIRICAL ARTICLE

The Temporal Dynamics of the Pretesting and Prequestioning Effects

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Making guesses about unfamiliar information, followed by correct answer feedback (*pretesting*) or subsequent learning activities such as reading (*prequestioning*), can enhance memory. Across two online experiments, we investigated the temporal dynamics of these memory benefits—known as the *pretesting effect* and *prequestioning effect*, respectively—relative to nonguessing approaches (i.e., reading, studying, or no-activity) among adult learners over retention intervals ranging from 5 min to 30 days. Analysis of variance and power-function fits revealed that the magnitude of both effects varied over time: Memory improvements were most evident over the first week, with indications that the pretesting effect remained detectable for up to 2 weeks. Forgetting rates did not significantly differ across guessing and nonguessing approaches. These findings indicate that pretesting and prequestioning confer memory boosts that are most prominent in the days immediately after learning, fortifying retention for at least 1 week before attenuating over longer intervals.

General Audience Summary

Making guesses about unfamiliar information, followed by learning the correct answers, can improve long-term memory. This approach is termed *pretesting* when guessing is coupled with immediate correct answer feedback, and *prequestioning* when the correct answers are learned through subsequent activities such as reading a text passage or watching videos. The present study investigated these *pretesting* and *prequestioning effects* across retention intervals ranging from 5 min to 1 month. Strong effects were observed relative to nonguessing approaches, but their magnitude varied with the passage of time. Specifically, both effects remained detectable for at least 1 week and then declined afterward, although there were indications that the pretesting effect remained detectable for up to 2 weeks. Together, these findings suggest that pretesting and prequestioning generate substantial memory boosts that improve retention during the period immediately following learning, with those benefits persisting for an educationally relevant duration before declining over longer periods of time. Ultimately, these results underscore the practical value of guessing-based activities as tools for enhancing learning effectively during the critical period following initial instruction.

Keywords: pretesting, prequestioning, long-term memory, retention interval, forgetting

Supplemental materials: <https://doi.org/10.1037/mac0000302.supp>

Robert Nash served as action editor.

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Materials, data, and analysis code are available at the Open Science Framework via <https://osf.io/du2ea/>. The authors declare no conflicts of interest. This research was supported by a Faculty of Arts and Social Sciences grant to Steven C. Pan, funded through the National University of Singapore. The authors thank Michelle Kaku for logistical support during study preparation and data collection and Pragati Maheshwary for assistance during exploratory pilot testing.

Steven C. Pan played a lead role in conceptualization, funding acquisition, methodology, project administration, resources, supervision,

visualization, writing—original draft, and writing—review and editing and a supporting role in formal analysis. Min Huey Teo played a lead role in software and a supporting role in conceptualization, data curation, formal analysis, and writing—review and editing. Liwen Yu played a lead role in formal analysis and a supporting role in writing—original draft and writing—review and editing. Gayathri D. Naidu played a lead role in data curation and validation and a supporting role in writing—review and editing.

 The data are available at <https://osf.io/du2ea/>.

 The experimental materials are available at <https://osf.io/du2ea/>.

 The preregistered design is available at <https://aspredicted.org/wz76x7.pdf>.

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Making erroneous guesses about unfamiliar information, followed by learning the correct answers, can enhance memory (for reviews, see Kornell & Vaughn, 2016; Metcalfe, 2017; Pan & Carpenter, 2023). For example, Kornell et al. (2009) found that guessing the answers to obscure general-knowledge questions before receiving correct answer feedback enhanced recall of the answers more than studying question–answer pairs. Similarly, guessing the answers to questions about a text passage or lecture prior to viewing it can improve memory relative to viewing the same materials without any prior guessing (e.g., Richland et al., 2009). When coupled with immediate correct answer feedback, such guessing is commonly called *pretesting*, and when feedback is absent and the answers are instead learned through a text passage or a lecture, it is often called *prequestioning* (e.g., St. Hilaire et al., 2024), although both terms are sometimes used interchangeably. The resulting learning advantages relative to a nonguessing condition are commonly referred to as the *pretesting effect* and *prequestioning effect*, respectively (Pan & Carpenter, 2023).

Pretesting and Prequestioning Effects Over Time

The pedagogical utility of pretesting and prequestioning as effective learning tools is heightened by the apparent generalizability of their effects over such diverse materials as facts and paired-associate words (e.g., Kornell et al., 2009), word-picture translations (Chua & Pan, 2026), text passages (e.g., Richland et al., 2009), and both live and video lectures (Carpenter & Toftness, 2017; Soderstrom & Bjork, 2023), as well as with individuals of different ages and cognitive abilities (Pan et al., 2025; Yu et al., 2025). Less is known, however, about the stability of their effects over longer and more practically meaningful retention intervals. Although a growing number of studies have reported significant pretesting and prequestioning effects after delays of at least 24 hr (e.g., Kornell, 2014, Experiment 3a; Pan et al., 2019), most evidence to date involves short delays of only a few minutes (Chan et al., 2018). Two recent meta-analyses, however, have found that retention interval does not moderate the prequestioning effect (King-Shepard et al., 2025; St. Hilaire et al., 2024), suggesting that the available evidence does not show systematic variation in effect magnitude over time. Relatedly, these meta-analyses have estimated the magnitude of the prequestioning effect as Hedges's $g = 0.54$ – 0.66 , which could be considered relatively large in terms of pedagogical interventions (Kraft, 2020), and more moderate-sized effects have been reported in some studies of pretesting in classrooms (e.g., a Cohen's d of 0.38 in Soderstrom & Bjork, 2023).

The few studies of pretesting and prequestioning that have manipulated retention interval directly provide further insights. Using Spanish paired-associate words, Mera et al. (2025; Experiment 1) found no significant difference in the magnitude of the pretesting effect at immediate and 24-hr retention intervals. Likewise, in a study using an expository text, Overoye et al. (2021; Experiment 2) found no significant difference in prequestioning effect magnitude after retention intervals of 5 min and 1 week. Although these studies suggest stability of both effects over time, the observed effect sizes were smaller numerically at longer delays (e.g., Mera et al. reported Cohen's d values of 1.29 and 0.69 at short and long delays, respectively, both of which could be characterized as medium-to-large to large effects under conventional and more recent interpretations; see Kraft, 2020).

In contrast, Kliegl et al. (2022), among the first investigations to examine pretesting and prequestioning effects across more than two experimentally manipulated retention intervals, found that the magnitude of both effects can *increase* over time. Experiment 1 in their study used paired-associate words, compared pretesting versus reading, and assessed memory after 1, 10, and 30 min; Experiment 2 used a historical text passage, compared prequestioning versus a reading-only condition, and assessed memory after 1 min, 30 min, and 1 week. In both experiments, a significant interaction between learning method and retention interval was observed. That interaction reflected progressively larger pretesting and prequestioning effects—from $d = 0.95$ (1 min) to 1.78 (30 min) in the case of pretesting and from $d = 0.89$ (1 min) to 1.89 (1 week) for prequestioning—across longer delays. On the basis of these results, Kliegl et al. concluded that the effects come to full fruition after extended delays.

Enhancing Encoding and Retarding Forgetting

The foregoing findings can be considered in light of major theoretical accounts of the pretesting and prequestioning effects. According to the three-stage framework of Pan and Carpenter (2023), both effects arise from cognitive processes engaged during guessing, the learning of correct answers, or both. The processes that are most responsible for the effects remain to be determined. Some theorized processes may initialize during the act of guessing—such as increased attention (Pan et al., 2020), curiosity (Potts et al., 2019), generation of an error signal (Seabrooke et al., 2021), and activation of relevant prior knowledge (Pashler et al., 2007)—and may primarily influence encoding. If so, then pretesting and prequestioning may yield higher initial memory performance without necessarily altering forgetting rates. In contrast, other theoretical processes that appear to depend critically on the moment when the correct answer is learned, such as the linkage of mediators between cues and targets (Carpenter, 2011), semantic network elaboration (Carpenter, 2009), search set formation and refinement (Grimaldi & Karpicke, 2012), and the formation of integrated memories for erroneous guesses and correct answers (Wahlheim & Jacoby, 2013), posit that qualitatively different memories are formed. If so, these processes may influence forgetting by altering the durability or accessibility of memory traces, rather than initial encoding strength. Overall, given that theoretical accounts appear to differ in their predictions about whether pretesting and prequestioning primarily influence initial memory strength, subsequent forgetting, or both, examining these effects across retention intervals should be informative.

As previously noted, the premise that pretesting and prequestioning retard forgetting rates is substantiated by at least one study (Kliegl et al., 2022), but other studies are more equivocal (e.g., King-Shepard et al., 2025; Mera et al., 2025; Overoye et al., 2021; St. Hilaire et al., 2024). Yet among these studies, retention interval has only been investigated in a fairly limited manner, with most experiments examining only a few time points and seldom extending beyond delays of 1 week. At present, the evidence available supports only broad directional inferences and does not yet permit detailed characterization or formal modeling of the functional form of forgetting following pretesting and prequestioning.

The Present Study

The two experiments reported in this article were designed to elucidate the temporal dynamics of the pretesting and

prequestioning effects. Experiment 1 focused on the former and Experiment 2 on the latter, with both using educationally relevant stimuli that are highly representative of the respective research literatures on each effect (factual and text passage materials, respectively), measuring learning across practically meaningful retention intervals spanning minutes to several weeks, and comparing performance against a potentially competitive control condition (i.e., reading, studying). Each experiment manipulated the delay between learning and a subsequent criterial test across a broader and more finely segmented range of retention intervals than in prior work, which has often focused on very short delays—from 5 min to 14 days in Experiment 1 and up to 1 month in Experiment 2—allowing for a more precise characterization of how both effects evolve over time.

Crucially, the design of both experiments facilitated analyses using power-function modeling alongside conventional analysis of variance approaches (Carpenter et al., 2008; Nickl & Bäuml, 2023), allowing for quantitative estimation of both initial performance differences and forgetting rates across conditions. The design also enabled a comprehensive examination of the extent to which the finding that pretesting and prequestioning effect magnitude increases as a function of retention interval generalizes. Ultimately, the present study provided a basis for distinguishing whether pretesting and prequestioning primarily affect initial learning strength, the rate of forgetting, or both.

Experiment 1

Experiment 1 investigated the temporal dynamics of the pretesting effect across retention intervals ranging from 5 min to 2 weeks. Participants learned a set of factual statements in an online setting via pretesting or reading and were randomly assigned to take a retention test after one of five delay intervals. Power functions were fit to retention test data in order to investigate the memory advantages following pretesting and the relative rates of forgetting for pretesting versus reading.

Method

Transparency and Openness

This article reports, for both experiments, how we determined our sample size, all data exclusions, all manipulations, and all measures. This experiment was not preregistered. All materials, data, and analysis code for the full study are archived at the Open Science Framework (OSF) via <https://osf.io/du2ea/>.

Design

Experiment 1 featured a 2 (learning method: pretesting vs. reading) $\times$ 5 (retention interval: 5-min vs. 1-day vs. 3-day vs. 7-day vs. 14-day) design. Learning method was manipulated within-subjects, and retention interval was manipulated between-subjects.

Participants

The target sample size of 36 participants per retention interval was based on prior studies of similar design that investigated the effects of different learning methods on long-term retention using analysis of variance (ANOVA) and other approaches (e.g., Kliegl et al.,

2022; see also Nickl & Bäuml, 2023; Toppino & Cohen, 2009). These studies reported effect sizes in the range $\eta_p^2 \approx 0.05$ – 0.21 for the interaction between learning method and retention interval. For the present design, featuring two learning conditions and five retention intervals, an a priori power analysis conducted using the *Superpower* package in R (Lakens & Caldwell, 2021) indicated that 180 participants, or 36 participants per retention interval, would exceed 90% power to detect a medium-sized interaction effect ($\eta_p^2 \approx 0.10$; Cohen's $f \approx 0.33$) at $\alpha = .05$. That simulation modeled two divergent forgetting curves where pretesting effect magnitude increased across retention intervals, reminiscent of the findings reported in Kliegl et al. (2022).

Two hundred twenty-seven participants were recruited from the research platform Prolific Academic. All participants had to be from an English-speaking country (i.e., Australia, Canada, New Zealand, the United Kingdom, or the United States), between 21 and 45 years old, fluent in English, and have a $\geq 90\%$ approval rate on the platform. Participants were not eligible to take part if they reported a diagnosis of mild cognitive impairment (MCI), dementia, dyslexia, or autism spectrum disorder. They were compensated in accordance with the Prolific minimum payment rate of U.S.\$8 per hour; participants assigned to the 5-min retention interval received U.S.\$3.34 following completion, whereas participants assigned to the remaining retention intervals received U.S.\$3.63 in total, consisting of U.S.\$1.86 for the first session, U.S.\$0.91 for the second session, and an additional U.S.\$0.91 bonus for completing it within 12 hr of the scheduled release time. The entire study was conducted with ethics approval obtained at the university with which the authors are affiliated. All participants gave informed consent before participation and were treated in accordance with the Declaration of Helsinki.

Prior to formal data analysis, data from six participants were excluded due to evidence of off-task browser activity, as detected via the TaskMaster add-on for Qualtrics (Permut et al., 2019). Participants were deemed off-task if more than 5 of the 8 s allotted per trial were spent off-task on at least three items during the learning phase. Data from 19 participants were excluded because they did not attempt the retention test within 24 hr or completed it after the 24-hr deadline, and data from six participants were excluded because they left six or more response boxes blank during the learning phase.

The final sample of 196 participants (5-min group, $n = 39$, 1-day group, $n = 37$, 3-day group, $n = 39$, 7-day group, $n = 39$, 14-day group, $n = 42$) had a mean age of 32.9 years and was 57% female. Ninety-five percent of these participants completed the retention test on time or within 12 hr, and all did so within 24 hr. Forty-eight percent of participants were from the United Kingdom, 21% were from the United States, and 31% were from the remaining eligible countries; 57% of the participants were White, 18% were Asian, and 25% were from other ethnic groups or declined to provide ethnicity information. Forty-three percent of these participants had completed an undergraduate degree; 32% had completed a postgraduate degree; and the remaining 25% had completed secondary education or held other nondegree qualifications.

Materials

The materials consisted of 32 factual statements drawn from Pan and Siau (2026), originally developed by converting word pairs from the Nelson et al. (1998) database into brief sentences. Compared

to the weakly associated word pairs used in Kliegl et al. (2022; Experiment 1), these statements were somewhat more semantically complex, although they shared the characteristic of linking two concepts in a concise format. They could also be categorized into such domains as biology and history and as such could be considered to be educationally relevant. Each statement linked a subject word with another word that defined or otherwise described it (e.g., “Pepsi contains caffeine”). Pretesting on these statements had previously been shown to yield significant pretesting effects relative to a reading control as measured on retention tests conducted after delays of 5-min and 24-hr (Cohen’s $d = 1.64$ and 0.45 , respectively, in Pan & Siau, 2026; the effect sizes were larger at the shorter delay). The materials are publicly available at the OSF archive for this study.

The stimulus set also included fill-in-the-blank versions of each statement in which the descriptive word was replaced with a blank (e.g., “Pepsi contains _____”). These versions were used during pretesting and on the retention tests. The 32 statements were split into two 16-item lists for counterbalancing purposes: During the training phase, each participant learned the items from one list using pretesting (in fill-in-the-blank format) and the items from the other list using reading; all 32 statements were then tested on the retention test in fill-in-the-blank format.

Procedure

All participants completed the experiment remotely on their personal laptop or desktop computer using an internet browser. The experiment comprised up to three phases: the learning phase, a distractor task phase (only for the 5-min group), and the retention test phase. Participants were randomly assigned via the Prolific platform to one of the retention interval conditions, and the study settings on the platform ensured that each participant could enroll in the experiment only once. All participants were told that the experiment would involve up to two separate sessions: The 5-min group completed all phases within a single session, after which they were informed that their participation was complete, and it was unnecessary to return for a second session, whereas all other groups completed the learning phase in the first session and the retention test in the second session.

Learning Phase. At the start of the experiment, participants read instructions explaining that they would encounter two types of statements: Incomplete statements in which they were to guess the missing word, and complete statements to read. To ensure that these instructions were understood, they completed a sample guessing trial and one attention check question before beginning the actual learning phase.

Participants then viewed the 32 factual statements presented in random order, with counterbalanced assignment of facts to pretesting or reading as described previously. For pretesting, participants saw a fill-in-the-blank version of a fact and had 8 s to guess and type the missing word into a provided textbox. They were encouraged to respond even if unsure and to avoid leaving the textbox empty. “I don’t know” responses were permitted. After 8 s, the complete statement appeared with the correct answer, accompanied by the instructions, “Read the complete fact, which includes the correct answer.” Correct answer feedback remained visible for 8 s before the next trial. For reading, intact statements were presented for 8 s each, with participants instructed to read the presented information.

It should be noted that the implementation of pretesting and reading in this experiment equated total exposure time to the correct answer (e.g., “caffeine”) in both conditions, but not total trial duration, which may have favored the pretesting condition. Differing trial time arrangements have been explored in the pretesting literature (e.g., Kornell et al., 2009), and equating total trial time by providing additional exposure to correct answers in the reading condition can reduce, but not necessarily reverse, the pretesting effect. Indeed, in Kliegl et al. (2022; Experiment 1), both the pretesting and reading conditions were allotted 10 s per trial, and significant pretesting effects were still observed.

Distractor Task. Participants in the 5-min group completed a distractor task immediately after the learning phase to minimize rehearsal. This task consisted of five 1-min open-ended prompts (e.g., listing celebrities or household items), during which participants typed out as many items as possible. Participants in all other groups skipped the distractor task and were instructed to wait for a message from the researchers, sent through the Prolific platform, which contained the link for the second session of the experiment.

Retention Test. For all participants except those in the 5-min group, the link to the retention test was provided to participants at the exact time corresponding to each group’s assigned retention interval (1 day, 3 days, 7 days, or 14 days), measured from completion of Part 1. Participants received the link via a Prolific message and were instructed to complete the second part at the assigned time; access remained open for up to 24 hr, with a bonus incentivizing completion within 12 hr. Those in the 5-min group simply proceeded directly to the retention test after completing the distractor task.

During the retention test, all 32 factual statements from the learning phase were presented in fill-in-the-blank format with one word missing. Participants were required to recall the missing word and type it into a provided text box. Each test trial was self-paced, and no feedback was provided. Backtracking to previous items was also disabled. Once participants had completed the retention test, they were debriefed and compensated for their participation.

Scoring and Data Analysis

Participants’ responses during pretesting and on the retention test were manually scored by a human rater. To be counted as correct, a given response had to match the correct answer exactly. However, minor spelling errors were allowed (i.e., “makeup” vs. “makeup”) provided that the intended response was unambiguous and that the misspelling did not form another valid word with a different meaning. For example, responses were scored as correct when minor misspellings still clearly reflected the intended answer (e.g., “lump” for “lumps”), whereas responses were scored as incorrect when the response formed a different valid word with a distinct meaning (e.g., “humps” instead of “lumps”). As each response involved a relatively simple, single-word answer, requiring an exact match was unlikely to disadvantage participants; prior research indicates that allowing for minor spelling variations does not meaningfully change scoring outcomes in similar test-enhanced learning paradigms (e.g., Pan et al., 2015). To determine the reliability of the scoring procedure, a second independent rater scored a randomly chosen subset of 152 criterial test responses using the same procedures, and no disagreements were observed among that subset (Cohen’s $\kappa = 1.0$).

Our analysis plan involved two approaches: an ANOVA to examine the effects of condition and retention interval, and a power-function analysis to model forgetting over time. First, a mixed-design ANOVA was conducted with retention interval (5 min, 1 day, 3 days, 7 days, 14 days) as the between-subjects factor and learning condition (pretesting and reading) as the within-subjects factor. Second, forgetting over time was modeled using the three-parameter power function, $r(t) = a(1 + ct)^{-b}$ (Rubin & Wenzel, 1996; see also Wickelgren, 1974; Wixted & Carpenter, 2007), which is described further below. The function was fit to recall performance using group-level data, with time expressed in days since the end of the learning phase. Parameters were estimated using maximum likelihood methods, and model fit was evaluated by comparing the likelihood of the power-function model with that of a baseline model that treated recall at each retention interval as independent binomial outcomes. Likelihood-ratio tests produced approximate χ^2 statistics indicating whether the power function provided an adequate description of the observed recall rates. Model fit was additionally summarized using R^2 , computed as 1 minus the ratio of squared error between observed and model-predicted recall proportions and the total variance in observed proportions at the group level.

Before testing individual conditions, we first estimated a common scaling parameter c across conditions, following previous approaches (Carpenter et al., 2008; Nickl & Bäuml, 2023; Siler & Benjamin, 2020). Specifically, we compared a restricted model in which parameters a (initial performance) and b (forgetting rate) were allowed to differ across conditions, but the scaling parameter c was constrained to be the same, with a more general model in which all parameters varied freely. Because the restricted model did not significantly reduce model fit relative to the general model, $\chi^2(1) = 1.49$, $p = .222$, the shared estimate of c ($c = 0.48$) was retained in subsequent analyses.

Using the common scaling parameter, we then evaluated the fit of the two-parameter power function for each condition separately, allowing parameters a and b to vary freely. To test whether conditions differed in initial performance or forgetting rate, we compared models in which parameters were allowed to vary across conditions with restricted models in which either a or b was constrained to be equal. These comparisons were conducted using likelihood-ratio tests. All model fitting procedures were conducted in R (Version 4.4.3) using the *optim()* function with the Nelder-Mead optimization algorithm.

Following precedent in the pretesting literature (e.g., Kliegl et al., 2022), analyses were first performed on retention test data where, in the pretesting condition, all items guessed correctly during the learning phase by a given participant were removed for that participant. Doing so eliminated the items for which that participant presumably had prior knowledge (although it should be noted that no such removals were possible for the reading condition). To address the potential consequences of this item removal, we repeated the analyses on retention data including all items, including those guessed correctly during the learning phase, with both sets of analyses reported in this article.

Results

Learning Phase Performance

During pretesting, across the entire sample, participants correctly guessed 10.1% of the answers to the fill-in-the-blank questions. As noted previously, analyses of retention test performance were

initially conducted after removing, for each participant, any items they had correctly guessed during pretesting.

Retention Test Performance

Analysis of Variance. The mixed-design ANOVA revealed a significant main effect of retention interval, $F(4, 191) = 56.63$, $MSE = 0.06$, $p < .001$, $\eta_p^2 = 0.54$, indicating that recall performance differed across time. There was also a significant main effect of learning method, $F(1, 191) = 94.90$, $MSE = 0.02$, $p < .001$, $\eta_p^2 = 0.33$, showing higher recall in the pretesting versus reading conditions. Importantly, the interaction between learning method and retention interval was significant, $F(4, 191) = 9.03$, $MSE = 0.02$, $p < .001$, $\eta_p^2 = 0.16$, suggesting that the pretesting effect varied across time.

Post hoc comparisons with Holm correction revealed that recall performance in the pretesting condition was significantly higher than in the reading condition at short and intermediate delays, specifically at 5 min, 1 day, and 3 days, whereas the difference was not significant at 7 days or 14 days (see Table 1). These results suggest that the pretesting effect emerged shortly after learning and persisted up to 3 days but was no longer evident at longer retention intervals.

We repeated the ANOVA including items that participants had correctly guessed during pretesting. The overall pattern of results remained unchanged, with the interaction between learning method and retention interval remaining significant, $F(4, 191) = 5.61$, $MSE = 0.02$, $p < .001$, $\eta_p^2 = 0.11$. Further, the pretesting effect remained significant at all retention intervals—from 5 min to 14-days—although its magnitude decreased over time (see Table 1).

Power Function Analysis. The recall data for the pretesting and reading conditions were modeled using a three-parameter power function. The function provided an adequate fit in both conditions, as indicated by nonsignificant goodness-of-fit tests, $\chi^2(3) \leq 6.68$, $p \geq .083$ (Table 2). The model captured the majority of variance in recall performance, with R^2 values $\geq .976$ across conditions.

To further characterize the pretesting effect, we performed likelihood-ratio tests comparing the parameters a (initial performance) and b (forgetting rate) between the pretesting and reading conditions. Parameter a differed significantly between conditions, $\chi^2(1) = 99.50$, $p < .001$ ($\Delta a = .23$, 95% CI [0.19, 0.28]), suggesting higher initial recall performance in the pretesting versus reading conditions. Parameter b did not significantly differ between conditions, $\chi^2(1) = 2.73$, $p = .098$ ($\Delta b = 0.08$, 95% CI [−0.02, 0.18]), indicating no reliable differences in rates of forgetting between the pretesting and reading conditions. Together, these results provided evidence that pretesting enhanced recall performance, with no reliable differences in estimated forgetting rates. The forgetting curves are presented in Figure 1.

We repeated the power-function analysis including items that participants had correctly guessed during pretesting. The pattern of results remained unchanged. The function with a common scaling parameter ($c = 0.57$) again provided an adequate fit, $\chi^2(3) \leq 6.12$, $p \geq .106$, with R^2 values $\geq .979$. Parameter a remained significantly higher in the pretesting condition, $\chi^2(1) = 117.92$, $p < .001$ ($\Delta a = .24$, 95% CI [0.20, 0.28]), whereas parameter b did not differ between conditions, $\chi^2(1) = 0.77$, $p = .381$ ($\Delta b = -0.04$, 95% CI [−0.11, 0.05]).

Discussion

Experiment 1 revealed that pretesting effect magnitude varies across retention intervals ranging from 5-min to 14 days. Pretesting

Table 1
Experiment 1 Post Hoc Comparisons of Pretesting and Reading Across Retention Intervals

Retention interval	Pretesting <i>M</i> (<i>SE</i>)	Reading <i>M</i> (<i>SE</i>)	Pretesting—reading <i>b</i> (<i>SE</i>)	<i>p</i>	Cohen's <i>d</i> (95% CI)
Excluding correctly guessed items					
5-min	0.82 (0.03)	0.60 (0.04)	0.21 (0.03)	<.001	1.04 [0.71, 1.36]
1-day	0.62 (0.03)	0.42 (0.04)	0.20 (0.03)	<.001	1.17 [0.83, 1.50]
3-day	0.49 (0.04)	0.36 (0.03)	0.13 (0.03)	<.001	0.77 [0.45, 1.10]
7-day	0.30 (0.03)	0.26 (0.03)	0.04 (0.03)	.148	0.24 [−0.08, 0.57]
14-day	0.20 (0.02)	0.17 (0.02)	0.03 (0.03)	.247	0.19 [−0.12, 0.51]
Including correctly guessed items					
5-min	0.83 (0.02)	0.60 (0.04)	0.23 (0.03)	<.001	1.13 [0.80, 1.45]
1-day	0.65 (0.03)	0.42 (0.04)	0.23 (0.03)	<.001	1.35 [1.02, 1.68]
3-day	0.53 (0.04)	0.36 (0.03)	0.18 (0.03)	<.001	1.00 [0.68, 1.33]
7-day	0.34 (0.03)	0.26 (0.03)	0.09 (0.03)	.003	0.50 [0.17, 0.82]
14-day	0.27 (0.02)	0.17 (0.02)	0.10 (0.03)	<.001	0.58 [0.27, 0.89]

Note. Significant values appear in bold. Post hoc comparisons are Holm-corrected. Cohen's *d* is computed from post hoc comparisons and indexes the standardized mean difference between pretesting and reading at each retention interval. *b* = unstandardized mean difference in terms of proportion correct; *SE* = standard error; CI = confidence interval.

generally supported better retention than reading, with the strongest effects emerging between 5-min and 3 days ($ds \geq 0.77$) and declining at longer intervals. Overall, these patterns suggest a dynamic time course for the effect: initially quite strong in the days following learning, then decreasing in observed magnitude rather than showing stability or growth over time. These patterns should be interpreted as reflecting different summaries of the same data, with the ANOVA reflecting performance across discrete retention bins and the power-function analysis modeling performance continuously over time.

Power function analyses further elucidated the forgetting dynamics of the pretesting effect. Pretesting yielded higher initial recall performance than reading, whereas no reliable differences in forgetting rates were observed. Together, the power-function analyses suggest that the pretesting effect is driven mainly by an initial recall performance advantage, with limited evidence for differences in estimated forgetting rates.

Experiment 2

Experiment 2 investigated the temporal dynamics of the prequestioning effect. It compared a prequestioning condition in which online learners attempted to answer questions prior to reading a text passage, a studying condition in which informative statements were read before the passage, and a passage-only control condition that involved no specific learning activities before the passage. Having these three conditions allowed the prequestioning effect to be measured against a control condition that engaged in a potentially

competitive activity, similar to Experiment 1, as well as another condition that did not, as in some prior research (e.g., Kliegl et al., 2022, Experiment 2; Pressley et al., 1990). By including both conditions, we sought to isolate the effect with greater precision. In particular, the studying condition featured exposure to relevant information, similar to the prequestioning condition, but removed the guessing component, thereby serving as a more stringent comparison than a passage-only control condition. Each participant was randomly assigned to one of six retention intervals ranging from 5 min to 30 days, allowing the fitting of power functions that, as in Experiment 1, assessed both initial memory advantages and relative rates of forgetting across conditions.

Method

Transparency and Openness

The experiment design, sample size target, and analysis plan were preregistered at <https://aspredicted.org/wz76x7.pdf>. Experiment 2 was preregistered prior to data collection as a planned extension of the study design first implemented in Experiment 1.

Design

Experiment 2 featured a 3 (learning method: prequestioning vs. studying vs. no-activity) $\times$ 6 (retention interval: 5-min vs. 1-day vs. 3-day vs. 7-day vs. 14-day vs. 30-day) design. Learning method was manipulated within-subjects, and retention interval was manipulated between-subjects.

Participants

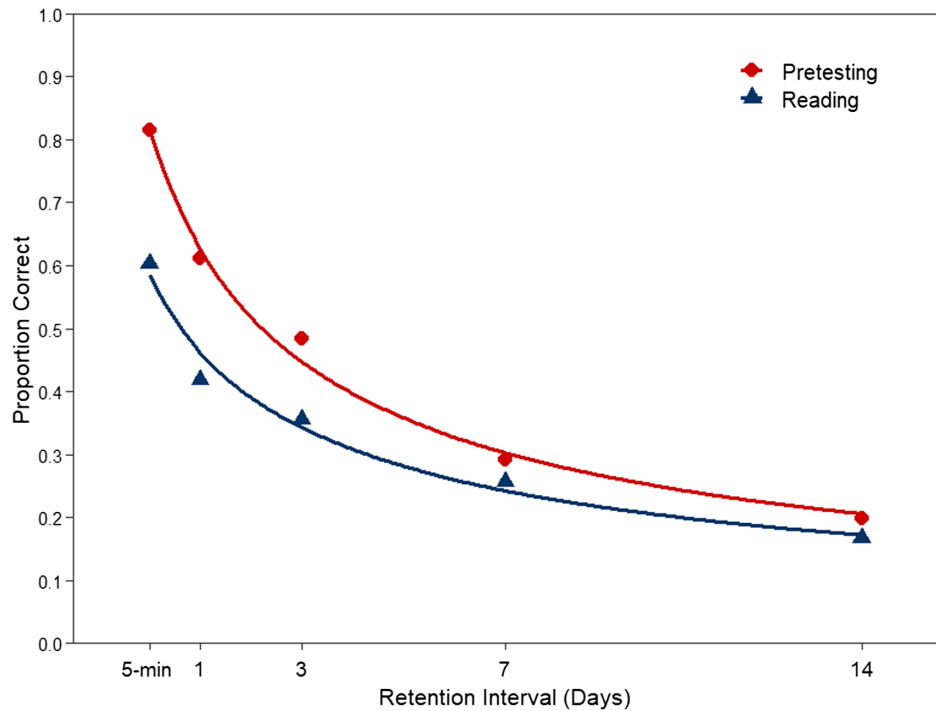
As in Experiment 1, we set a sample size target of 36 participants per retention interval, based on effect sizes for Delay $\times$ Practice-Type interactions reported in prior research (e.g., Kliegl et al., 2022). For the current design, which featured three within-subjects learning conditions and six between-subjects retention intervals, an a priori power analysis conducted in R using the *Superpower* package

Table 2
Experiment 1 Power Function Parameter Estimates and Model Fit Across Conditions

Condition	<i>a</i>	<i>b</i>	$\chi^2(3)$	<i>p</i>	<i>R</i> ²
Pretesting	0.82	0.67	3.86	.277	.993
Reading	0.58	0.60	6.68	.083	.976

Note. *a* = initial performance parameter; *b* = forgetting rate parameter.

Figure 1
Experiment 1 Forgetting Curves for Pretesting and Reading



Note. Mean proportion of correct recall displayed as a function of learning condition and retention interval. Points represent observed group means, whereas the solid lines represent best-fitting power functions. Items guessed correctly were removed from the pretesting condition. See the online article for the color version of this figure.

indicated that 216 participants, or 36 participants per retention interval, would exceed 90% power to detect a medium-sized interaction effect ($\eta_p^2 \approx 0.08$, Cohen's $f \approx 0.30$) at $\alpha = .05$ in a scenario with three divergent forgetting curves and increasing performance differences across longer retention intervals (cf. Kliegl et al., 2022). In this scenario, prequestioning produced the highest performance, followed by studying and then the passage-only condition. This ordering was deemed plausible based on differences in exposure to relevant information prior to reading the passage and the additional requirement to attempt to generate answers in the prequestioning condition, relative to the other conditions, with potential implications for variation across retention intervals.

It should be noted that the addition of the 30-day retention interval reflected an extension of the design following Experiment 1, for which the longest retention interval was 14 days. Experiment 1 was designed to remain within platform payment requirements (i.e., completion of payments within 21 days) and addressed concerns about potential attrition at longer delays. The successful completion of that experiment with minimal attrition prompted us to extend the retention intervals in Experiment 2 to capture longer term memory dynamics.

Two hundred seventy-six participants were recruited from the research platform Prolific Academic. All participants had to meet the same age, language ability, and approval rate criteria as in Experiment 1. In Experiment 2, however, participants from the United States were excluded to minimize the possibility of prior familiarity with the text passage, which addressed a U.S. historical figure. Only participants from the United Kingdom, Australia, Canada, and New Zealand were

eligible. Similar to Experiment 1, exclusion criteria included self-reported diagnoses of MCI, dementia, dyslexia, or autism spectrum disorder. In line with Prolific minimum payment rates, participants assigned to the 5-min retention interval received U.S.\$3.21 following completion, whereas participants assigned to the remaining retention intervals received U.S.\$3.08 in total, consisting of U.S.\$1.88 for the first session, U.S.\$0.60 for the second session, and an additional bonus of approximately U.S.\$0.60 for completion of the second session within 12 hr of being granted access to it.

Prior to formal data analysis, data from four participants were excluded due to evidence of off-task browser activity (i.e., more than 90 of the 300 s allotted were spent off-task during the text passage reading phase), and data from 31 participants were excluded because they did not attempt the retention test within 24 hr or completed it after the 24-hr deadline. The final sample of 241 participants (5-min group, $n = 40$, 1-day group, $n = 39$, 3-day group, $n = 38$, 7-day group, $n = 39$, 14-day group, $n = 40$, 30-day group, $n = 45$) had a mean age of 34.1 years and was 54% female. Ninety-three percent of these participants completed the retention test on time or within 12 hr, and all did so within less than 24 hr. Seventy-six percent of these participants were from the United Kingdom, and 24% were from the remaining eligible countries; 74% of the participants were White, 10% were Asian, and 16% were from other ethnic groups or declined to provide ethnicity information. Forty-five percent of these participants had completed an undergraduate degree; 29% had completed a postgraduate degree; and the remaining 26% had completed secondary education or held other nondegree qualifications.

Materials

The materials included an encyclopedic text passage, 15 short-answer questions derived from the passage, and 15 statements also derived from the passage. The passage was a 268-word text about former U.S. President Woodrow Wilson, adapted from the version originally used in Kliegl et al. (2022; Experiment 2), who in turn drew from the English Wikipedia article on the subject (https://en.wikipedia.org/wiki/Woodrow_Wilson). The passage was translated from German into English. Prequestioning on this passage had previously been shown to yield significant prequestioning effects relative to a no-activity control as measured on retention tests conducted after delays of 1-min, 30-min, and 7-days (Cohen's $d = 1.27, 1.84, \text{ and } 2.68$, respectively, in Kliegl et al., 2022), with larger effect sizes as the delay increased.

We adapted the questions (e.g., "In which year did Woodrow Wilson's presidency begin?") from Kliegl et al.'s stimulus materials and created the statements by transforming those questions into declarative sentences (e.g., "Woodrow Wilson's presidency began in a specific year early in the 20th century."). Neither the prequestions nor the statements included the correct answer; thus, in both conditions, participants had to consult the subsequent passage to learn the correct information, thereby equating exposure to the correct answers across conditions and paralleling the use of learning-objective control statements in prior prequestioning research (e.g., Pan et al., 2026; Sana et al., 2020; Schweppe et al., 2025).

In Kliegl et al. (2022), participants received eight of 16 total questions as prequestions prior to reading the text passage, whereas the remaining eight items were assigned to the passage-only condition; all 16 questions were then presented on the retention test. In the present experiment, which compared prequestioning with both a time-matched control activity (reading statements) and a passage-only condition, we used 15 questions/statements divided into three sets of five items each. Each participant received five prequestions and five statements prior to reading the passage and then completed a retention test containing all 15 short-answer questions. The assignment of each set to condition was counterbalanced over participants.

Procedure

As in Experiment 1, all participants completed the experiment remotely on their personal laptop or desktop computer and using an internet browser. The session structure, retention interval assignments, and platform procedures were identical to those described for Experiment 1. Participants in the 5-min group completed all phases within a single session, whereas all other groups completed the learning phase in the first session and the retention test in a second session.

Learning Phase. Participants first read instructions stating that they would encounter two types of factual statements, some requiring guessing the answer ("Please read this question carefully and make your best guess even if unsure") and others being intact statements that required only reading. It was emphasized that not knowing the correct answer to the questions was acceptable at this stage. As in Experiment 1, participants completed a sample guessing trial and one attention check before beginning the actual learning phase.

Participants then viewed five prequestions and five statements presented in random order, with counterbalanced assignment as described previously. For each prequestion, participants were required to type their answer into a provided textbox. "I don't know" responses were

permitted. Unlike Experiment 1, the prequestions and statements were self-paced rather than timed, and no feedback was provided, which replicates the approach taken in Kliegl et al. (2022). Doing so avoided participants from being cutoff while responding, consistent with prior prequestioning studies that have used self-pacing (e.g., Carpenter & Toftness, 2017) but not with other studies that imposed fixed time limits (e.g., de Lima & Jaeger, 2020). Descriptive statistics for the time taken during prequestioning and reading statements are reported in the Results section. Immediately afterward, the passage was presented, and participants were allotted exactly 5 min to read it.

Distractor Task and Retention Test. As in Experiment 1, participants in the 5-min group completed a distractor task, which was identical to the version used in the prior experiment, prior to taking the retention tests. They were then told that it was unnecessary to return for a second session. Participants in all other groups were informed to return for a second session and were provided with a link to that session in the same manner as in Experiment 1; the link remained accessible for up to 24 hr, with a bonus incentivizing completion within 12 hr of receipt. During the retention test, all 15 questions were presented in short-answer format and in random order, with participants required to type the answer to each question into a provided textbox. As in the prior experiment, each test trial was self-paced, feedback was not provided, and backtracking to previous items was disabled. After completing the retention test, each participant was debriefed and compensated for their participation.

Scoring and Data Analysis

Because participants' responses frequently involved proper names or multiword answers, prequestion and retention test responses were hand-scored by a human rater who was blind to experimental condition. A response was counted as correct if it contained all words of the correct answer or at least one complete component of a multiword proper noun (e.g., for the correct answer "William Taft," the responses "William" and "Taft" were counted as correct). Minor misspellings and extraneous words were not penalized provided that the aforementioned conditions were met. The reliability of the scoring procedure was verified by a second independent rater who scored a randomly chosen subset of 158 critical test responses using the same procedures, yielding perfect agreement with the primary rater (Cohen's $\kappa = 1.0$).

Data analysis followed the same approach as Experiment 1. For the ANOVA, retention interval (5 min, 1 day, 3 days, 7 days, 14 days, 30 days) was treated as a between-subjects factor, and learning condition (prequestioning, statements, and no-activity) was treated as a within-subjects factor. Greenhouse–Geisser corrected F tests were used when sphericity was violated. For the power-function analysis, a restricted model with a shared scaling parameter c across conditions did not significantly reduce fit compared with the general model, $\chi^2(2) = 1.55, p = .460$, so $c = 1.02$ was used in all subsequent analyses. Also, as in Experiment 1, analyses were conducted both on retention test data where items guessed correctly during prequestioning were removed and where those items were not removed.

Results

Learning Phase Performance

During prequestioning, across the entire sample, participants correctly guessed 10.0% of the answers to the short-answer questions.

Based on page-level metadata from the Qualtrics platform, the time spent was $M = 17.10$ s ($SE = 0.48$) for guessing (prequestioning) trials and $M = 7.66$ s ($SE = 0.35$) for studying trials. These values reflect increased time-on-task among the latter trial type, presumably due to the added duration needed for guessing and typing.

Retention Test Performance

Analysis of Variance. The mixed-design ANOVA revealed a significant main effect of retention interval, $F(5, 235) = 29.14$, $MSE = 0.13$, $p < .001$, $\eta_p^2 = 0.38$, indicating that recall performance differed across time. The main effect of learning method was also significant, $F(1.96, 461.63) = 34.59$, $MSE = 0.04$, $p < .001$, $\eta_p^2 = 0.13$, showing higher recall in the prequestioning condition compared with the studying and no-activity conditions. The interaction between retention interval and learning method was significant, $F(9.82, 461.63) = 3.11$, $MSE = 0.04$, $p < .001$, $\eta_p^2 = 0.06$, indicating that the prequestioning effect varied across time.

Post hoc comparisons with Holm correction revealed that the prequestioning condition outperformed the studying and no-activity conditions significantly at 5-min, 1 day, 3 days, and 7 days, but not at 14 days or 30 days (see Tables 3 and 4). These results suggest that prequestioning effect emerged shortly after learning and persisted for up to 1 week but was no longer strongly evident at even longer retention intervals.

We repeated the ANOVA including items that participants had correctly guessed during prequestioning. The overall pattern of results remained unchanged, with the interaction between learning method and retention interval remaining significant, $F(9.95, 467.62) = 3.01$, $p = .001$, $\eta_p^2 = 0.06$. Post hoc comparisons showed the same overall pattern: prequestioning outperformed the studying and no-activity conditions significantly at 5-min, 1 day, 3 days, and 7 days, but not at 14 days or 30 days (see Tables 3 and 4).

Power Function Analysis. The three-parameter power function provided an adequate fit in the studying and no-activity

conditions, as indicated by nonsignificant goodness-of-fit tests, $\chi^2(4) \leq 4.62$, $p \geq .329$ (Table 5). In contrast, the goodness-of-fit test was significant for the prequestioning condition, suggesting some deviation from the power-law curve in this condition, $\chi^2(4) = 11.72$, $p = .020$. Nevertheless, the models captured the majority of variance in recall performance, with R^2 values $\geq .942$ across conditions.

To evaluate the prequestioning effect, likelihood-ratio tests were conducted to compare the parameters a (initial performance) and b (forgetting rate) between each pair of conditions. Parameter a differed significantly between prequestioning and studying conditions, $\chi^2(1) = 12.86$, $p < .001$, $\Delta a = .15$, 95% CI [0.07, 0.22], between the prequestioning and no-activity conditions, $\chi^2(1) = 46.15$, $p < .001$, $\Delta a = .29$, 95% CI [0.22, 0.38], and between the studying and no-activity conditions, $\chi^2(1) = 10.37$, $p = .001$, $\Delta a = .14$, 95% CI [0.06, 0.24]. Hence, prequestioning yielded higher initial recall performance than studying and no-activity. In contrast, parameter b did not differ significantly between any pair of conditions, $\chi^2(1) \leq 0.99$, $p \geq .321$, $\Delta b \leq 0.05$, all 95% CIs included zero, indicating no reliable differences in forgetting rates across conditions. These results provided evidence that prequestioning enhanced recall performance without altering the forgetting rate. The forgetting curves are shown in Figure 2.

We repeated the analysis including items that participants had correctly guessed during prequestioning. The pattern of results remained unchanged. The function with a common scaling parameter ($c = 0.63$) provided an adequate fit for the studying and no-activity condition, but not for the prequestioning condition, $\chi^2(4) = 13.18$, $p = .010$, although the model still accounted for substantial variance ($R^2 = .935$). Parameter a in the prequestioning condition remained significantly higher than in the studying condition, $\chi^2(1) = 17.94$, $p < .001$, $\Delta a = .17$, 95% CI [0.09, 0.26], and the no-activity condition, $\chi^2(1) = 56.93$, $p < .001$, $\Delta a = .30$, 95% CI [0.22, 0.36]. In contrast, parameter b did not differ across conditions, $\chi^2(1) \leq 0.93$, $p \geq .334$, $\Delta b \leq 0.05$, all 95% CIs included zero, again indicating comparable forgetting rates.

Table 3
Experiment 2 Post Hoc Comparisons of Prequestioning and Studying Across Retention Intervals

Retention interval	Prequestioning <i>M</i> (<i>SE</i>)	Studying <i>M</i> (<i>SE</i>)	Prequestioning—studying <i>b</i> (<i>SE</i>)	<i>p</i>	Cohen's <i>d</i> (95% CI)
Excluding correctly guessed items					
5-min	0.79 (0.04)	0.67 (0.05)	0.12 (0.05)	.015	0.38 [0.06, 0.70]
1-day	0.54 (0.04)	0.43 (0.05)	0.11 (0.05)	.047	0.39 [0.06, 0.71]
3-day	0.55 (0.05)	0.36 (0.05)	0.19 (0.05)	<.001	0.57 [0.24, 0.90]
7-day	0.41 (0.05)	0.28 (0.04)	0.13 (0.05)	.013	0.40 [0.08, 0.72]
14-day	0.27 (0.04)	0.25 (0.04)	0.03 (0.05)	.837	0.10 [−0.22, 0.42]
30-day	0.15 (0.03)	0.17 (0.04)	−0.02 (0.05)	>.999	−0.07 [−0.37, 0.23]
Including correctly guessed items					
5-min	0.79 (0.04)	0.67 (0.05)	0.12 (0.05)	.01	0.38 [0.06, 0.70]
1-day	0.57 (0.04)	0.43 (0.05)	0.15 (0.05)	.003	0.58 [0.25, 0.90]
3-day	0.58 (0.05)	0.36 (0.05)	0.23 (0.05)	<.001	0.72 [0.40, 1.05]
7-day	0.45 (0.05)	0.28 (0.04)	0.17 (0.05)	<.001	0.53 [0.21, 0.85]
14-day	0.32 (0.04)	0.25 (0.04)	0.07 (0.05)	.260	0.25 [−0.07, 0.57]
30-day	0.19 (0.03)	0.17 (0.04)	0.01 (0.04)	.891	0.05 [−0.25, 0.35]

Note. Significant values appear in bold. Post hoc comparisons are Holm-corrected. Cohen's d is computed from post hoc comparisons and indexes the standardized mean difference between prequestioning and no-activity at each retention interval. b = unstandardized mean difference in terms of proportion correct; SE = standard error; CI = confidence interval.

Table 4*Experiment 2 Post Hoc Comparisons of Prequestioning and No-Activity Across Retention Intervals*

	Prequestioning	No activity	Prequestioning—no-activity		
Retention interval	<i>M</i> (<i>SE</i>)	<i>M</i> (<i>SE</i>)	<i>b</i> (<i>SE</i>)	<i>p</i>	Cohen's <i>d</i> (95% CI)
Excluding correctly guessed items					
5-min	0.79 (0.04)	0.51 (0.05)	0.27 (0.05)	<.001	0.98 [0.79, 1.16]
1-day	0.54 (0.04)	0.36 (0.04)	0.17 (0.05)	.001	0.55 [0.36, 0.73]
3-day	0.55 (0.05)	0.29 (0.05)	0.26 (0.05)	<.001	0.80 [0.61, 0.98]
7-day	0.41 (0.05)	0.26 (0.04)	0.15 (0.05)	.006	0.50 [0.31, 0.68]
14-day	0.27 (0.04)	0.21 (0.04)	0.06 (0.05)	.552	0.20 [0.02, 0.39]
30-day	0.15 (0.03)	0.14 (0.03)	0.01 (0.04)	>.999	0.05 [−0.12, 0.22]
Including correctly guessed items					
5-min	0.79 (0.04)	0.51 (0.05)	0.28 (0.05)	<.001	0.99 [0.81, 1.17]
1-day	0.57 (0.04)	0.36 (0.04)	0.21 (0.05)	<.001	0.71 [0.52, 0.89]
3-day	0.58 (0.05)	0.29 (0.05)	0.29 (0.05)	<.001	0.98 [0.80, 1.17]
7-day	0.45 (0.05)	0.26 (0.04)	0.18 (0.05)	<.001	0.62 [0.44, 0.80]
14-day	0.32 (0.04)	0.21 (0.04)	0.11 (0.05)	.061	0.36 [0.18, 0.54]
30-day	0.19 (0.03)	0.14 (0.03)	0.04 (0.04)	.89	0.19 [0.01, 0.36]

Note. Significant values appear in bold. Post hoc comparisons are Holm-corrected. Cohen's *d* is computed from post hoc comparisons and indexes the standardized mean difference between prequestioning and no-activity at each retention interval. *b* = unstandardized mean difference in terms of proportion correct; *SE* = standard error; CI = confidence interval.

Discussion

Experiment 2 revealed that the prequestioning effect varies across retention intervals ranging from 5 min to 30 days. Prequestioning generally supported better retention than studying and a no-activity condition, with significant advantages evident between 5 min ($d = 0.38$) and 1 week ($ds \geq 0.40$), and reduced effects at longer delays. As with the pretesting effect, these patterns suggest a strong prequestioning effect in the days immediately after learning, followed by a reduction in the observed effect over time rather than stability or growth.

Power function analyses revealed a similar pattern, with prequestioning yielding higher initial recall performance than studying or no-activity, but no reliable differences in forgetting rates across conditions. Inclusion or exclusion of correctly guessed items did not meaningfully alter the results. Together, these analyses suggest that the prequestioning effect is driven by an initial recall performance advantage, with limited evidence for differences in estimated forgetting rates over extended retention intervals, alongside some deviation from the assumed functional form in the prequestioning condition.

General Discussion

The present study investigated the temporal dynamics of the pretesting and prequestioning effects across retention intervals ranging from 5 min to 30 days. In each experiment, the magnitude of

the observed effects varied across time, a pattern substantiated by ANOVA evidence. In Experiment 1, the pretesting effect was relatively large at 5 min ($d \geq 1.0$) and remained substantial at 3 days ($d \geq 0.77$) but was smaller at 1 week ($d \geq 0.24$) and at longer intervals. Similarly, in Experiment 2, the prequestioning effect was evident at 5 min ($d = 0.38$), highest at 3 days ($d \geq 0.57$), remained significant at 1 week ($d \geq 0.40$), but was not reliably detected beyond 2 weeks.

Power function analyses provided further converging evidence. In both experiments, the initial performance parameter differed between pretesting or prequestioning and their respective control condition(s), whereas the forgetting rate parameter did not reliably differ across conditions. Hence, both effects appear to be driven by differences in initial recall performance rather than differences in forgetting rates. This finding helps clarify the ANOVA findings: Although the absolute difference in recall between the pretesting/prequestioning and respective control condition(s) decreased over time, this reduction is consistent with the overall decline in recall rather than a faster rate of forgetting for pretesting or prequestioning conditions. Moreover, declining levels of overall recall at longer retention intervals may constrain the observable separation between conditions and reduce the precision with which these differences can be detected. As evident in the forgetting curves in Figures 1 and 2, pretesting and prequestioning produced an initial recall advantage that became smaller as overall recall declined, with forgetting trajectories that remained comparable across conditions.

Overall, this study reveals similar temporal trajectories for the pretesting and prequestioning effects. Both effects are characterized by an early boost in recall that is evident within 5 min and detectable at 1 week, with the absolute magnitude of the recall advantage decreasing over time. Neither method appears to reduce forgetting rates.

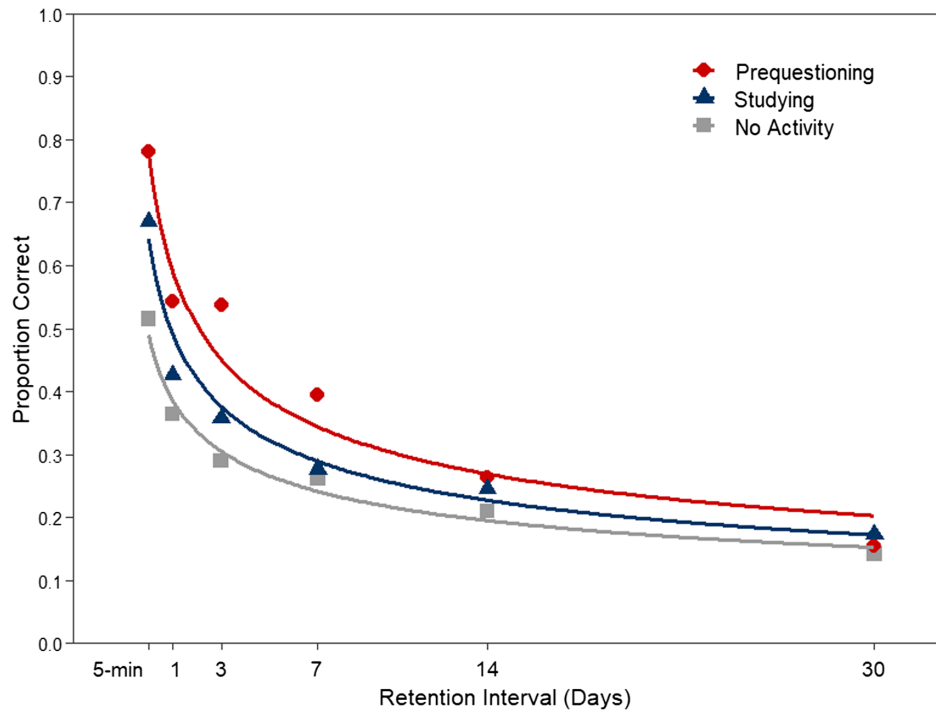
Implications for Pretesting and Prequestioning Research

The present findings align with the results of Mera et al. (2025) and Overoye et al. (2021), where effects decreased at longer delays. Although Kliegl et al. (2022) suggested that floor effects might explain

Table 5*Experiment 2 Power Function Parameter Estimates and Model Fit Across Conditions*

Condition	<i>a</i>	<i>b</i>	$\chi^2(4)$	<i>p</i>	<i>R</i> ²
Prequestioning	0.78	0.39	11.72	.020	.942
Studying	0.64	0.39	4.62	.329	.965
No activity	0.50	0.35	1.98	.739	.976

Note. Significant values appear in bold. *a* = initial performance parameter; *b* = forgetting rate parameter.

Figure 2*Experiment 2 Forgetting Curves for Prequestioning, Study, and No Activity*

Note. Mean proportion of correct recall displayed as a function of learning condition and retention interval. Points represent observed group means, whereas the solid lines represent best-fitting power functions. Items guessed correctly were removed from the prequestioning condition. See the online article for the color version of this figure.

Overoye et al.'s results at 7 days, our data demonstrate similar patterns without floor-level performance. Regarding the meta-analytic finding that retention interval does not moderate the prequestioning effect (King-Shepard et al., 2025; St. Hilaire et al., 2024; cf. Chan et al., 2018), our results are consistent insofar that no linear or monotonic relationship was observed. Within-experiment manipulations of retention interval (unlike most studies in those meta-analyses, where retention interval was potentially confounded with other design differences), however, revealed that pretesting and prequestioning effects can vary across retention intervals in a nonmonotonic manner.

When considering the present results in relation to Kliegl et al. (2022), a key consideration is that we sampled a different and wider range of retention intervals. The progressive effect size increases that they observed for the pretesting effect across 1-min ($d = 0.95$), 10-min ($d = 1.69$), and 30-min ($d = 1.78$) intervals are not directly comparable to our results for pretesting across different timescales. Their finding of larger effects for prequestioning versus no-activity at 1-week ($d = 2.68$) versus 1-min ($d = 1.27$) or 30-min ($d = 1.84$), however, differs directionally from our findings for prequestioning at 5-min ($d = 0.98$) and 1-week ($d = 0.50$). Given our use of the same (albeit translated) stimulus materials and many of the same procedures, this discrepancy implies that the prequestioning effect is more sensitive to retention interval and other boundary conditions than previously thought.

Regarding the possibility that the pretesting (and prequestioning) effects come to full fruition after prolonged retention intervals, the

present data are inconsistent with a progressive strengthening of either effect over time. That possibility further raises the issue of whether any effect size increases would persist or reverse over time; the present data are more consistent with attenuation. Notably, however, our prequestioning results imply an effect size peak around 3 days, broadly resembling the trajectory reported by Kliegl et al. It is also notable that their reported effects at longer delays were quite large ($d \approx 2.0$) relative to other studies using similar retention intervals (e.g., Latimier et al., 2019; Richland et al., 2009), raising the prospect of different trajectories under stronger performance levels.

Theoretically, the present results are consistent with the premise that cognitive processes engaged during guessing enhance initial encoding of target information (e.g., Pan et al., 2020; see also Potts et al., 2019; Seabrooke et al., 2021) without necessarily altering longer-term retention dynamics. If the resulting memories were qualitatively different (e.g., mediator formation), we might expect reduced forgetting as well. The power-function results also contrast with the literature on posttesting (i.e., retrieval practice), where studies manipulating retention interval to fit power functions have found that it reduces forgetting rates without reliably affecting initial performance (e.g., Carpenter et al., 2008; Nickl & Bäuml, 2023; see also Siler & Benjamin, 2020), the reverse pattern to that observed here. This divergence may reflect different mechanisms engaged by the two approaches, with pretesting and prequestioning primarily enhancing initial encoding and posttesting strengthening consolidation (Dong et al., 2026).

Study Limitations and Future Directions

As perhaps the first study to investigate pretesting and prequestioning using both retention interval manipulation and curve-fitting, this work establishes an initial benchmark for future research. Several limitations, however, warrant consideration. First, the temporal resolution was relatively coarse and the stimulus materials were fairly limited. Although prior knowledge was low (as evidenced by the ~10% mean guessing rate), the materials were arguably less challenging than other stimuli such as foreign language word pairs. Second, although analyses of delayed retention test completion times and error types did not suggest alternative explanations for the findings (see [Supplemental Results](#)), future studies could equate time across conditions (e.g., [Kornell et al., 2009](#)) or systematically manipulate exposure durations. Such work might also examine individual differences ([Pan et al., 2025](#)), improved instantiations of guessing and/or feedback ([Khin & Pan, 2026](#)), other strategies that participants may have engaged in alongside guessing ([Rivers et al., 2026](#)), and learning in classroom contexts ([Soderstrom & Bjork, 2023](#)). Generalizability across contexts, including classroom environments and online samples where pretesting and prequestioning effects may be more difficult to obtain, also warrants further investigation.

Another potential concern involves the use of aggregated group-level data in nonlinear curve fitting. To address this issue, in alignment with recommendations from prior research ([Anderson & Tweney, 1997](#); [Nickl & Bäuml, 2023](#)), we repeated the power-function analyses using a log-based (geometric) transformation of the observed recall rates. As detailed in the [Supplemental Results](#), these analyses yielded very similar results to those involving the original fitting approach. More generally, however, it should be noted that interpretations of parameter differences depend on the assumed power-function form, and alternative models of retention may yield different characterizations of change over time.

Practical Implications

The present study confirms the potency of pretesting and prequestioning in enhancing the encoding of factual and text-based materials, adding to a growing body of evidence demonstrating their pedagogical utility ([Pan & Carpenter, 2023](#)). Our findings suggest that these benefits are most evident during the period immediately following learning and remain detectable over a significant, albeit not interminable, duration. Hence, instructors and learners may benefit most from prioritizing these methods to bolster retention during critically important periods of initial learning, such as bridging the gap between an early-week introductory lecture and a subsequent lab or tutorial. To maintain performance beyond that window, however, additional review or practice may be necessary. Ultimately, the present results provide a clearer roadmap for using pretesting and prequestioning where they are likeliest to offer the most significant and reliable gains for learners.

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Received May 11, 2026

Revision received July 12, 2026

Accepted August 2, 2026 ■

Journal of Applied Research in Memory and Cognition

The Temporal Dynamics of the Pretesting and Prequestioning Effects

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Online First Publication, September 21, 2026. <https://dx.doi.org/10.1037/mac0000302>

CITATION

Pan, S. C., Teo, M. H., Yu, L., & Naidu, G. D. (2026). The temporal dynamics of the pretesting and prequestioning effects. *Journal of Applied Research in Memory and Cognition*. Advance online publication. <https://dx.doi.org/10.1037/mac0000302>